WORKED SOLUTION

STEP II 2014, Question 1

Pure • Euclidean geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q1

The key idea

Walk from A to B along the zig-zag $A \to P \to Q \to B$. The three legs all have length x, and together they must cover a horizontal distance of exactly 1 and climb a net height of 0. Resolving horizontally and vertically gives $x\cos\theta$ and $x\sin\theta$, and $\cos^2\theta + \sin^2\theta = 1$ removes the unknown angle. Everything after that is a quadratic in x and its discriminant.

Solution

Put A at the origin and B at $(1, 0)$, with C above AB. Since P is at distance x from A along the ray from A through C, and Q is at distance x from B along the ray from B through C,

$$P = (x\cos\alpha,\ x\sin\alpha), \qquad Q = (1 - x\cos\beta,\ x\sin\beta).$$

Part (i)

Horizontally. The horizontal components of $\overrightarrow{AP}$, $\overrightarrow{PQ}$ and $\overrightarrow{QB}$ are $x\cos\alpha$, $x\cos\theta$ and $x\cos\beta$, and they add up to $AB = 1$. So

$$x\cos\theta = 1 - x\cos\alpha - x\cos\beta,$$

which is the first result. (In coordinates: it is the difference of the x-coordinates of Q and P.)

Vertically. P is at height $x\sin\alpha$ and Q at height $x\sin\beta$, so

$$x\sin\theta = x\sin\beta - x\sin\alpha.$$

(This is never negative: $\alpha \le \beta < 180^\circ - \alpha$ because the angles of the triangle add up to less than 180°, and on that interval $\sin\beta \ge \sin\alpha$. So Q is never lower than P.)

Eliminating θ. Squaring and adding, and using $\cos^2\theta + \sin^2\theta = 1$,

$$x^2 = (1 - x(\cos\alpha + \cos\beta))^2 + x^2(\sin\beta - \sin\alpha)^2.$$

Expanding the right-hand side, the coefficient of x^2 is

$$\begin{aligned}(\cos\alpha+\cos\beta)^2+(\sin\beta-\sin\alpha)^2 &= (\cos^2\alpha+\sin^2\alpha)+(\cos^2\beta+\sin^2\beta)\\ &\quad+2(\cos\alpha\cos\beta-\sin\alpha\sin\beta)\\ &= 2+2\cos(\alpha+\beta).\end{aligned}$$

So $x^2 = 1-2(\cos\alpha+\cos\beta)x+(2+2\cos(\alpha+\beta))x^2$. Taking x^2 to the right-hand side gives

$$(1+2\cos(\alpha+\beta))x^2-2(\cos\alpha+\cos\beta)x+1=0, \qquad (*)$$

as required.

P and Q on the sides produced. If P lies on AC produced, it is still on the ray from A through C, at distance x from A, so its coordinates are still $(x\cos\alpha, x\sin\alpha)$. Likewise Q is still $(1-x\cos\beta, x\sin\beta)$. The derivation above used nothing else: the displacement from P to Q is

$$(1-x\cos\alpha-x\cos\beta,\ x\sin\beta-x\sin\alpha),$$

and $PQ^2 = x^2$ gives exactly the same equation $(*)$. (Now x may exceed AC and BC, and $1-x\cos\alpha-x\cos\beta$ may be negative, so θ may be obtuse. This doesn't matter, because the signs disappear on squaring.)

Part (ii)

The equation $(*)$ is linear exactly when $1+2\cos(\alpha+\beta)=0$, that is, $\cos(\alpha+\beta)=-\frac{1}{2}$. Since $0<\alpha+\beta<180^\circ$, the only possibility is

$$\alpha+\beta = 120^\circ.$$

Otherwise $(*)$ is a genuine quadratic, and a quarter of its discriminant is

$$\begin{aligned}\tfrac{1}{4}\Delta &= (\cos\alpha+\cos\beta)^2-(1+2\cos\alpha\cos\beta-2\sin\alpha\sin\beta)\\ &= \cos^2\alpha+\cos^2\beta-1+2\sin\alpha\sin\beta\\ &= 1-\sin^2\alpha-\sin^2\beta+2\sin\alpha\sin\beta\\ &= 1-(\sin\beta-\sin\alpha)^2,\end{aligned}$$

using $\cos^2\alpha = 1-\sin^2\alpha$ and $\cos^2\beta-1=-\sin^2\beta$.

? **Ask yourself** Why must $(\sin\beta-\sin\alpha)^2$ be less than 1?

Both angles lie strictly between 0° and 180°, so $0<\sin\alpha\le 1$ and $0<\sin\beta\le 1$. The difference of two numbers in $(0,1]$ lies strictly between -1 and 1. Hence $(\sin\beta-\sin\alpha)^2<1$, so $\Delta>0$ and $(*)$ has two distinct real roots.

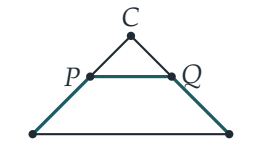
Part (iii)

(a) $\alpha=\beta=45^\circ$**.** Here $\cos(\alpha+\beta)=\cos 90^\circ=0$ and $\cos\alpha+\cos\beta=\sqrt{2}$, so $(*)$ is

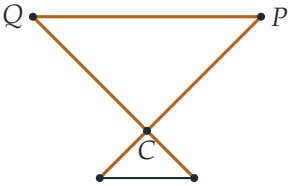
$$x^2-2\sqrt{2}\,x+1=0, \qquad x=\sqrt{2}\pm 1.$$

The triangle is right-angled and isosceles, with $C = (\frac{1}{2}, \frac{1}{2})$ and $AC = BC = \frac{1}{\sqrt{2}} \approx 0.707$.

- $x = \sqrt{2} - 1 \approx 0.414 < AC$: P and Q lie on the sides, at $P = (1 - \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}})$ and $Q = (\frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}})$. By symmetry PQ is parallel to AB, so $\theta = 0$. Check: $PQ = \sqrt{2} - 1 = x$.
- $x = \sqrt{2} + 1 \approx 2.414 > AC$: P and Q lie on AC produced and BC produced, at $P = (1 + \frac{1}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}})$ and $Q = (-\frac{1}{\sqrt{2}}, 1 + \frac{1}{\sqrt{2}})$. The two produced lines have crossed at C, so P is now to the *right* of Q. PQ is again horizontal, with length $1 + \sqrt{2} = x$.



$x = \sqrt{2} - 1$

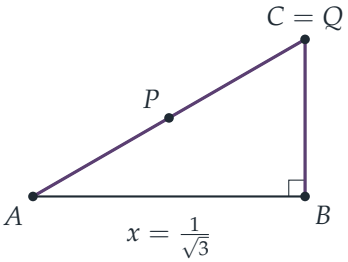


$x = \sqrt{2} + 1$

(b) $\alpha = 30°$**,** $\beta = 90°$**.** Now $\alpha + \beta = 120°$, so this is the linear case of part (ii). With $\cos\alpha + \cos\beta = \frac{\sqrt{3}}{2}$, $(*)$ becomes

$$-\sqrt{3}\,x + 1 = 0, \qquad x = \frac{1}{\sqrt{3}}.$$

The triangle has its right angle at B, with $BC = \tan 30° = \frac{1}{\sqrt{3}}$ and $AC = \frac{2}{\sqrt{3}}$. So $QB = BC$ means Q *is* C, and $AP = \frac{1}{\sqrt{3}} = \frac{1}{2}AC$ puts P at the midpoint of AC. Then $PQ = PC = \frac{1}{\sqrt{3}} = x$, as it should be. This is the only possibility.



$x = \frac{1}{\sqrt{3}}$

Where has the second root gone? If $\alpha + \beta = 120°$, the lines AC and BC meet at 60°. For very large x, the points P and Q far out along them are then almost exactly at distance x apart, forming a nearly equilateral triangle with C. As $\alpha + \beta \to 120°$, one root of the quadratic runs off to infinity.

Checks

- In (a), $(\sqrt{2}-1)(\sqrt{2}+1) = 1$, matching the product of the roots, $1/(1+2\cos 90^\circ) = 1$.
- For a random triangle, say $\alpha = 40^\circ$, $\beta = 70^\circ$, solve $(*)$, place P and Q and measure PQ: it equals x for both roots. We confirmed this by computer for many random triangles, to about 10^{-10}.
- In (b), $x = 1/\sqrt{3} \approx 0.577$ is exactly BC, and the midpoint of AC is $(\frac{1}{2}, \frac{1}{2\sqrt{3}})$, at distance $\frac{1}{\sqrt{3}}$ from $C = (1, \frac{1}{\sqrt{3}})$.

Common slips

- Getting the sign of the $\cos\beta$ term wrong: $\overrightarrow{QB}$ runs *towards* B, so its horizontal component is $+x\cos\beta$, even though Q is to the left of B.
- In part (ii), allowing $\alpha + \beta = 240^\circ$. The angles of a triangle add up to less than 180°.
- In (b), trying to use the quadratic formula when the leading coefficient is zero, or concluding that there is no solution.

Beyond STEP

Resolving a closed polygon of vectors into components, then eliminating an angle with $\cos^2 + \sin^2 = 1$, is a standard technique in kinematics and linkage problems: our zig-zag is a simple three-bar linkage. The root that "escapes to infinity" when the quadratic becomes linear is a first glimpse of projective geometry. There, parallel lines meet at a point at infinity, and a degree-two equation always has two roots once these are counted. Coxeter and Greitzer, *Geometry Revisited*, is a lovely way into classical geometry at this level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.